Unitary Representations

Group Actions on Measure Spaces

Let G be a topological group. The action of G on a measure space $(\Omega, \mathcal{M}, \mu)$ is called measurable if the map $G \times \Omega \to \Omega$ taking $(g, \omega) \mapsto g.\omega$ is measurable when G is equipped with the Borel σ-algebra.

We say the measure μ is invariant if $g_*\mu(A) = \mu(A)$ for all $A \in \mathcal{M}$. The measure is said to be quasi-invariant if μ and $g_*\mu$ are of the same measure class (or mutually absolutely continuous).

If $G \curvearrowright \Omega$ is a measurable group action on a σ-finite space, or G acts continuously on the locally compact space Ω with μ a regular Borel measure, then there exists a positive measurable function, the Radon–Nikodym derivative $\frac{dg_*\mu}{d\mu}$, on Ω, such that

$$\int f(g.\omega) \frac{dg_*\mu}{d\mu}(\omega)\, d\mu(\omega) = \int f(\omega)\, d\mu(\omega).$$

The function $c_\mu \colon G \times \Omega \to [0, \infty]$ given by $c_\mu(g, \omega) = \frac{dg_*\mu}{d\mu}(\omega)$ is measurable and satisfies the cocycle relation

$$c_\mu(g_1 g_2, \omega) = c_\mu(g_1, g_2.\omega) c_\mu(g_2, \omega)$$

for all $g_1, g_2 \in G$ and a.e. $\omega \in \Omega$.

Define for each $g \in G$ an operator π_g on $L^2(\Omega, \mu)$ by

$$\pi_g f(\omega) = c_\mu\left(g^{-1}, \omega\right)^{1/2} f\left(g^{-1}.\omega\right)$$

for all $f \in L^2(\Omega, \mu)$ and $\omega \in \Omega$. Thus, the operator π_g is unitary, and the cocycle relation gives that $\pi \colon G \to \mathcal{U}\left(L^2(\Omega, \mu)\right)$ is a group homomorphism.

Lemma: Suppose G is a locally compact group, and $\mathscr{H}$ is a separable Hilbert space. Let $\pi \colon G \to \mathcal{U}(\mathscr{H})$ is a homomorphism from G to the unitary group of $\mathscr{H}$. If the function $g \mapsto \langle \pi(g)\xi, \xi \rangle$ is measurable for all $\xi \in \mathscr{H}$, then π is SOT-continuous.

Proof. Let ξ be a vector in $\mathscr{H}$. It suffices to show that $g \mapsto \pi(g)\xi$ is continuous at e.

Let $\varepsilon > 0$ and $A = \{g \in G : \|\pi(g)\xi - \xi\| < \varepsilon/2\}$. Then, A is measurable since

$$A = \left\{g \in G : 2\,\mathrm{Re}(\langle \pi(g)\xi, \xi \rangle) > 2\|\xi\|^2 - \varepsilon^2/4\right\},$$

and $A = A^{-1}$ with $A^2 = AA^{-1} \subseteq \{g \in G : \|\pi(g)\xi - \xi\| < \varepsilon\}$.

The subset $\pi(G)\xi$ is separable since $\mathscr{H}$ is separable. Thus, there is a sequence $(g_n)_n$ in G such that $(\pi(g_n)\xi)_n$ is dense in $\pi(G)\xi$. For any $g \in G$, we have

$$\begin{aligned} \left\|\pi\left(g_n^{-1} g\right)\xi - \xi\right\| &= \|\pi(g)\xi - \pi(g_n)\xi\| \\ &< \varepsilon/2, \end{aligned}$$

so that $G = \bigcup_n g_n A$. Since the Haar measure of G is not zero, the Haar measure of A is not zero, so $A^2 = AA^{-1}$ is a neighborhood of e. □

Proposition: Suppose G is a σ-compact locally compact group, and (Ω, μ) is a σ-finite measurable space such that $L^2(\Omega, \mu)$ is separable, and let $G \curvearrowright \Omega$ be a measure class preserving action. Then, $\pi \colon G \to L^2(\Omega, \mu)$ defined by $\pi_g f(\omega) = c\left(g^{-1}, \omega\right)^{1/2} f\left(g^{-1}.\omega\right)$ is a unitary representation of G on $L^2(\Omega, \mu)$.

Proof. Since g is σ-compact, the matrix coefficient realization

$$g \mapsto \left\langle \pi_g f_1, f_2 \right\rangle = \int c_\mu\left(g^{-1}, \omega\right)^{1/2} f_1\left(g^{-1}.\omega\right) \overline{f_2(\omega)}\, d\mu(\omega)$$

is measurable by Fubini's theorem, hence SOT continuous. □

Kernels and Functions of Positive Type

Let X be a topological space

Definition: A kernel of positive type on X is a continuous function $\Phi : X \times X \to \mathbb{C}$ such that for any $n \in \mathbb{N}$, any elements $x_1, \dots, x_n \in X$, and any $c_1, \dots, c_n \in \mathbb{C}$,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i \overline{c_j} \Phi\left(x_i, x_j\right) \geq 0.$$

Equivalently, the matrix $\left(\Phi\left(x_i, x_j\right)\right)_{i,j} \in \mathbb{M}_n$ is positive.

Proposition: If Φ is a kernel of positive type on X, then for all $x, y \in X$, one has

(i) $\Phi(y,x) = \overline{\Phi(x,y)}$;

(ii) $\left|\Phi(x,y)\right|^2 \leq \Phi(x,x)\Phi(y,y)$ (i.e., Cauchy–Schwarz).

Proof. When $n = 2$, $x_1 = x$, and $x_2 = y$, one observes that the matrix

$$A = \begin{pmatrix} \Phi(x,x) & \Phi(x,y) \\ \Phi(y,x) & \Phi(y,y) \end{pmatrix}$$

is positive, so it is hermitian with positive diagonal values satisfying $\Phi(x,x)\Phi(y,y) - \Phi(y,x)\Phi(x,y) \geq 0$. □

The canonical example is the following. If $f : X \to \mathscr{H}$ is a continuous map into a Hilbert space, then

$$\Phi(x,y) = \langle f(x), f(y) \rangle$$

is a kernel of positive type on X.

Theorem (GNS for Kernels of Positive Type)**:** Let Φ be a kernel of positive type on a topological space X. Then, there exists a Hilbert space $\mathscr{H}$ and a continuous map $f : X \to \mathscr{H}$ such that

(i) $\Phi(x,y) = \langle f(x), f(y) \rangle$;

(ii) $\overline{\operatorname{span}}\{f(x) : x \in X\} = \mathscr{H}$.

This Hilbert space is unique up to isomorphism.

Proof. For every $x \in X$, we let $\Phi_x(y) = \Phi(x,y)$. These are each continuous functions. Let $V = \operatorname{span}\{\Phi_x : x \in X\} \subseteq C(X)$. Given two elements of V

$$\varphi = \sum_{i=1}^{m} a_i \Phi_{x_i}$$
$$\psi = \sum_{j=1}^{n} b_j \Phi_{x_j},$$

we write

$$\begin{aligned} \langle \varphi, \psi \rangle &= \sum_{i=1}^{m} \sum_{j=1}^{n} a_i \overline{b_j} \Phi\left(x_i, x_j\right) \\ &= \sum_{j=1}^{n} \overline{b_j} \varphi\left(x_j\right) \\ &= \sum_{i=1}^{m} a_i \overline{\psi(x_i)}. \end{aligned}$$

The form $\langle \varphi, \psi \rangle$ is a positive sesquilinear form on V. We claim it is definite. Notice that $\langle \varphi, \Phi_x \rangle = \varphi(x)$, and by Cauchy–Schwarz,

$$\left|\varphi(x)\right|^2 \leq \Phi(x,x)\langle \varphi, \varphi \rangle.$$

Completing V with respect to the induced norm yields the Hilbert space $(\mathscr{H}, \langle \cdot, \cdot \rangle)$.

If $f : X \to V$ takes $x \mapsto \Phi_x$, then $\langle f(x), f(y) \rangle = \Phi(x,y)$, and f is continuous since

$$\left\| f(x) - f(y) \right\|^2 = \Phi(x,x) + \Phi(y,y) - 2\,\mathrm{Re}(\Phi(x,y))$$

and Φ is continuous.

Now, suppose $(\mathscr{K}, g)$ is another pair satisfying (i) and (ii). Then,

$$\begin{aligned}
\left\| \sum_{i=1}^n \Phi_{x_i} \right\|^2 &= \sum_{i=1}^n \sum_{j=1}^n a_i \overline{a_j} \Phi\left(x_i, x_j\right) \\
&= \sum_{i=1}^n \sum_{j=1}^n a_i \overline{a_j} \left\langle g(x_i), g\left(x_j\right) \right\rangle \\
&= \left\| \sum_{i=1}^n a_i g(x_i) \right\|^2.
\end{aligned}$$

Thus, if we define the map $V \to \mathscr{K}$ by taking $\Phi_{x_i} \mapsto g(x_i)$ and extending linearly, we obtain an isometry $T : \mathscr{H} \to \mathscr{K}$. Since $T(V) = \mathrm{span}\{g(x) : x \in X\}$, which is dense in $\mathscr{K}$, we have that $T \circ f = g$. □

We note that, given any kernels of positive type Φ and Ψ, the following are also kernels of positive type:

(i) $\overline{\Phi}$;

(ii) $t\Phi$ for all $t \geq 0$;

(iii) $\Phi + \Psi$;

(iv) $\Phi\Psi$;

(v) the pointwise limit of a family of kernels of positive type, given that the limit is continuous.

Thus, the kernels of positive type are a convex cone in the vector space of continuous kernels on X that is closed under conjugation, pointwise product, and pointwise convergence.

Lemma: Let Φ be a kernel of positive type on X such that $\left|\Phi(x,y)\right| < 1$ for all $x, y \in X$. Then, $(1 - \Phi)^{-t}$ is a kernel of positive type for all $t \geq 0$.

Proof. The series

$$1 + \sum_{j=1}^{\infty} \frac{\prod_{k=0}^{j-1}(t+k)}{j!} \Phi(x,y)^j$$

converges to $(1 - \Phi(x,y))^{-t}$ for all $x, y \in X$ by the assumption that $\left|\Phi(x,y)\right| < 1$ for all $x, y \in X$. □

A sort of dual to the kernels of positive type is a kernel of conditionally negative type. They are related to each other by Schoenberg's Theorem.

Definition: A real-valued kernel $\Psi : X \times X \to \mathbb{R}$ is said to be of *conditionally negative type* if

(i) $\Psi(x,x) = 0$ for all $x \in X$;

(ii) $\Psi(x,y) = \Psi(y,x)$ for all $x, y \in X$;

(iii) for all elements $x_1, \dots, x_n$ and all real numbers $c_1, \dots, c_n$ with $\sum c_i = 0$, one has

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \Psi\left(x_i, x_j\right) \leq 0.$$

The canonical example is that of the map on a real Hilbert space

$$\Psi : \mathscr{H} \times \mathscr{H} \to \mathbb{R}$$
$$(\xi, \eta) \mapsto \left\|\xi - \eta\right\|^2.$$

One observes that for any $\xi_1, \dots, \xi_n \in \mathscr{H}$ and $c_1, \dots, c_n \in \mathbb{R}$ with $\sum c_i = 0$, we have

$$\begin{aligned}\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \Psi\left(\xi_i, \xi_j\right) &= \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \left\|\xi_i - \xi_j\right\|^2 \\ &= -2\left\|\sum_{i=1}^{n} c_i \xi_i\right\|^2.\end{aligned}$$

So, if X is a topological space and $\mathscr{H}$ is a real Hilbert space, with $f : X \to \mathscr{H}$ a continuous mapping, then the kernel $(x, y) \mapsto \left\|f(x) - f(y)\right\|^2$ is of conditionally negative type. Similar to how kernels of positive type arise, this is universal via a GNS construction.

Theorem (GNS for Kernels of Conditionally Negative Type)**:** Let Ψ be a kernel of conditionally negative type on X, and let $x_0 \in X$. There exists a real Hilbert space $\mathscr{H}$ and a continuous map $f : X \to \mathscr{H}$ such that

(i) $\Psi(x, y) = \left\|f(x) - f(y)\right\|^2$;

(ii) $\overline{\operatorname{span}}\{f(x) - f(x_0) : x \in X\} = \mathscr{H}$.

Moreover, this pair is unique up to affine isometry.

Proof. Let $V = \ker(\epsilon)$, where $\epsilon : \mathbb{R}[X] \to \mathbb{R}$ is the augmentation map

$$\sum_{i=1}^{m} c_i \delta_{x_i} \mapsto \sum_{i=1}^{m} c_i.$$

Let $\varphi = \sum_{i=1}^{m} a_i \delta_{x_i}$ and $\psi = \sum_{j=1}^{n} b_j \delta_{x_j}$ in V. Then, we let

$$\langle \varphi, \psi \rangle = -\frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_i b_j \Psi\left(x_i, x_j\right).$$

This defines a positive symmetric form on V. Set

$$N_\Psi = \{\varphi \in V : \langle \varphi, \varphi \rangle = 0\}.$$

By the Cauchy–Schwarz inequality, this is a linear subspace of V. Define

$$\langle [\varphi], [\psi] \rangle = \langle \varphi, \psi \rangle$$

on V/N_Ψ. Let $(\mathscr{H}, \langle \cdot, \cdot \rangle)$ be the Hilbert space completion of V/N_Ψ. Set $f : X \to H$ taking $x \mapsto \left[\delta_x - \delta_{x_0}\right]$. Then,

$$\left\|f(x) - f(y)\right\|^2 = \{si(x, y),$$

since $\Psi(x, x) = 0$ and $\Psi(y, y) = 0$ with $\Psi(y, x) = \Psi(x, y)$. Moreover, f is continuous since Ψ is continuous, so we obtain existence.

Let $(\mathscr{K}, g)$ be another such pair. We claim that the map

$$\left[\sum_{i=1}^{n} a_i \delta_{x_i}\right] \mapsto \sum_{i=1}^{n} a_i g(x_i)$$

extends to an affine isometry. Since $\sum_{i=1}^n a_i = 0$, it follows that

$$\begin{aligned}
\left\|\left[\sum_{i=1}^n a_i \delta_{x_i}\right]\right\|^2 &= \left\|\left[\sum_{i=1}^n a_i\left(\delta_{x_i} - \delta_{x_i}\right)\right]\right\|^2 \\
&= -\frac{1}{2}\sum_{i,j=1}^n a_i a_i\left(\Psi\left(x_i, x_j\right) - \Psi(x_i, x_0) - \Psi\left(x_j, x_0\right)\right) \\
&= -\frac{1}{2}\sum_{i,j=1}^n a_i a_j\left(\left\|g(x_i) - g\left(x_j\right)\right\|^2 - \|g(x_i) - g(x_0)\|^2 - \left\|g\left(x_j\right) - g(x_0)\right\|^2\right) \\
&= \sum_{i,j=1}^n a_i a_j\left\langle g(x_i) - g(x_0), g\left(x_j\right) - g(x_0)\right\rangle \\
&= \left\|\sum_{i=1}^n a_i (g(x_i) - g(x_0))\right\|^2 \\
&= \left\|\sum_{i=1}^n a_i g(x_i)\right\|^2.
\end{aligned}$$

Moreover, since $\operatorname{span}\{g(x) - g(x_0) : x \in X\}$ is dense in $\mathscr{K}$, it follows that T is surjective with $T \circ f = g$. □

Analogously to the case of kernels of positive type, there are certain inheritance properties for kernels of conditionally negative type (i.e., kernels of conditionally negative type form a convex cone and are closed in the topology of pointwise convergence on the space of continuous functions). Furthermore, if Φ is a real valued kernel of positive type, then $\Phi(x,x) - \Phi(x,y)$ is a kernel of conditionally negative type.

Lemma: Let X be a topological space, and let $\Psi : X \times X \to \mathbb{R}$ be a continuous kernel with $\Psi(x,x) = 0$ and $\Psi(y,x) = \Psi(x,y)$ for all $x, y \in X$. For a fixed $x_0 \in X$, let $\Phi : X \times X \to \mathbb{R}$ be defined by

$$\Phi(x,y) = \Psi(x,x_0) + \Psi(y,x_0) - \Psi(x,y).$$

Then, Ψ is a kernel of conditionally negative type if and only if Φ is a kernel of positive type.

Proof. Suppose Φ is a kernel of positive type. Then, for any $x_1, \ldots, x_n \in X$ and $c_1, \ldots, c_n \in \mathbb{R}$ with $\sum_{i=1}^n c_i = 0$, we have

$$\begin{aligned}
0 &\leq \sum_{i=1}^n \sum_{j=1}^n c_i c_j \Phi\left(x_i, x_j\right) \\
&= -\sum_{i=1}^n \sum_{j=1}^n c_i c_j \Psi\left(x_i, x_j\right),
\end{aligned}$$

showing that Ψ is of conditionally negative type.

Suppose Ψ is of conditionally negative type. Then, by the GNS for kernels of conditionally negative type, there is a real Hilbert space $\mathscr{K}$ and a continuous map $f : X \to \mathscr{K}$ such that $\Psi(x,y) = \left\|f(x) - f(y)\right\|^2$ for all $x, y \in X$. Then, we have

$$\begin{aligned}
\Phi(x,y) &= \Psi(x,x_0) + \Psi(y,x_0) - \Psi(x,y) \\
&= 2\langle f(x) - f(x_0), f(y) - f(x_0)\rangle
\end{aligned}$$

for all $x, y \in X$. Thus, for all $x_1, \ldots, x_n \in X$ and $c_1, \ldots, c_n \in \mathbb{R}$, we have

$$\sum_{i,j=1}^n c_i c_j \Phi\left(x_i, x_j\right) = 2\left\|\sum_{i=1}^n c_i (f(x_i) - f(x_0))\right\|^2$$

$$\geq 0.$$

Thus, since Ψ is symmetric the expression

$$\sum_{i,j=1}^{n} c_i \overline{c_j} \Phi\left(x_i, x_j\right) \geq 0$$

holds for all $x_1, \ldots, x_n \in X$ and all complex $c_1, \ldots, c_n$. Thus, Φ is of positive type. □

Theorem (Schoenberg's Theorem)**:** Let X be a topological space, $\Psi : X \times X \to \mathbb{R}$ a continuous kernel on X with $\Psi(x,x) = 0$ and $\Psi(y,x) = \Psi(x,y)$ for all $x, y \in X$. Then, the following are equivalent:

(i) Ψ is of conditionally negative type;

(ii) $\exp(-t\Psi)$ is of positive type for all $t \geq 0$.

Proof. Suppose $\exp(-t\Psi)$ is of positive type for every $t \geq 0$. Then, the kernel $1 - \exp(-t\Psi)$ is of conditionally negative type, and so is the pointwise limit

$$\Psi = \lim_{t \to 0} \frac{1 - \exp(-t\Psi)}{t}.$$

This gives the desired result.

Suppose now that Ψ is of conditionally negative type. It suffices to show that $\exp(-\Psi)$ is of positive type. Let $x_0 \in X$. Then, the kernel

$$\Phi(x,y) = \Psi(x,x_0) + \Psi(y,x_0) - \Psi(x,y)$$

is of positive type. Notice then that

$$\exp(-\Psi(x,y)) = \exp(\Phi(x,y)) \exp(-\Psi(x,x_0)) \exp(-\Psi(y,x_0)).$$

Since Φ is of positive type, it is also the case for $\Phi(x,y)^n$, hence $\exp(\Phi)$ is of positive type. Since

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i \overline{c_j} \exp(-\Psi(x_i,x_0)) \exp\left(-\Psi\left(x_j,x_0\right)\right) = \left| \sum_{i=1}^{n} c_i \exp(-\Psi(x_i,x_0)) \right|^2 \geq 0$$

for all $x_1, \ldots, x_n \in X$ and complex numbers $c_1, \ldots, c_n$. Thus, $\exp(-\Psi)$ is of positive type since it is the product of kernels of positive type. □

Functions of Positive Type

The GNS Construction for Representations

Definition: A function of positive type on a locally compact group G is a continuous function $\varphi : G \to \mathbb{C}$ such that the kernel $(g_1, g_2) \mapsto \varphi\left(g_2^{-1} g_1\right)$ is of positive type.

Proposition: Let φ be a function of positive type on G. Then, for all $g, h \in G$, we have

(i) $\overline{\varphi(g)} = \varphi\left(g^{-1}\right)$

(ii) $\left|\varphi(g)\right| \leq \varphi(e)$

(iii) $\left|\varphi(g) - \varphi(h)\right|^2 \leq 2\varphi(e)\left(\varphi(e) - \mathrm{Re}\left(\varphi\left(g^{-1}h\right)\right)\right)$.

Proof. Part (i) follows from the fact that

$$\begin{aligned}\varphi(g) &= \varphi((g)(e)) \\ &= \overline{\varphi((e)(g^{-1}))} \\ &= \overline{\varphi(g^{-1})},\end{aligned}$$

and part (ii) follows from

$$\begin{aligned}\left|\varphi(g)\right|^2 &= \left|\varphi((g)(e))\right| \\ &\leq \varphi\left(g^{-1}g\right)\varphi\left(e^{-1}e\right) \\ &= \varphi(e)^2.\end{aligned}$$

Finally, (iii) follows from the fact that for all $x, y, z \in \mathbb{C}$, we have

$$|x|^2 + \left|y\right|^2 + |z|^2 + 2\,\mathrm{Re}\left(\varphi\left(g^{-1}\right)\overline{y}x + \varphi\left(h^{-1}\right)\overline{z}x + \varphi\left(g^{-1}h\right)\overline{y}z\right) \geq 0$$

since φ is of positive type. When

$$\begin{aligned}x &= \left|\varphi(g) - \varphi(h)\right| \\ y &= -\frac{\left|\varphi(g) - \varphi(h)\right|}{\varphi(g) - \varphi(h)} \\ z &= \frac{\left|\varphi(g) - \varphi(h)\right|}{\varphi(g) - \varphi(h)},\end{aligned}$$

we have

$$\left|\varphi(g) - \varphi(h)\right|^2 + 2 + 2\,\mathrm{Re}\left(\frac{}{(-\varphi(g) + \varphi(h))}\frac{\left|\varphi(g) - \varphi(h)\right|^2}{\overline{\varphi(g) - \varphi(h)}} - \varphi\left(g^{-1}h\right)\right) \geq 0.$$

□

Proposition: Let $(\pi, \mathscr{H})$ be a unitary representation of G, and let ξ be a vector in $\mathscr{H}$. Then, the matrix coefficient $g \mapsto \langle \pi(g)\xi, \xi\rangle$ is a function of positive type.

Proof. Immediate from the discussed example. □

Definition: Let G be a locally compact group with Haar measure dx. Given functions f, g on G, the convolution $f * g$ is given by

$$\begin{aligned}f * g(x) &= \int f(xy)g\left(y^{-1}\right)dy \\ &= \int f(y)g\left(y^{-1}x\right)dy.\end{aligned}$$

Proposition: Every matrix coefficient of the regular representation of a locally compact group G vanishes at infinity.

Proof. We note that the matrix coefficients of the left regular representation of G are given by

$$\left\{f * \check{g} : f, g \in L^2(G)\right\},$$

where $\check{g}(x) = g\left(x^{-1}\right)$. Since $C_c(G) \subseteq L^2(G)$ is dense, there exist $f_n, g_n \in C_c(g)$ such that $\lim f_n = f$ and $\lim g_n = g$. So, $\lim f_n * \check{g}_n = f * \check{g}$ uniformly on G.

Thus, to show that $f * \check{g} \in C_0(G)$, we may analogously assume that $f, g \in C_c(G)$. If $K = \mathrm{supp}(f)$ and $L = \mathrm{supp}(g)$, then $\mathrm{supp}(f * \check{g}) \subseteq KL^{-1}$. Thus, $f * \check{g} \in C_c(G)$. □

Corollary: Let G be a non-compact locally compact group. Then, λ_G has no finite dimensional subrepresentation.

Proof. Let $(\pi, \mathscr{H})$ be a finite-dimensional unitary representation of G, and let $(v_i)_i$ be an orthonormal basis. It suffices to show that some matrix coefficient $\left\langle \pi(\cdot)v_i, v_j \right\rangle$ does not vanish at infinity.

Yet, observe that $\left(\left\langle \pi(g)v_i, v_j \right\rangle\right)_{i,j}$ is the matrix of $\pi(g)$ with respect to the basis $(v_i)_i$, so it is a unitary matrix. Thus, for any i, we have

$$\sum_j \left|\left\langle \pi(g)v_i, v_j \right\rangle\right|^2 = 1$$

for all $g \in G$. □

Definition: A unitary representation $(\pi, \mathscr{H})$ of G is cyclic if there exists $\xi \in \mathscr{H}$ such that $\overline{\text{span}}\left\{\pi_g \xi : g \in G\right\}$ is dense in $\mathscr{H}$. We call ξ a cyclic vector for π.

Theorem (GNS for Topological Groups)**:** Let φ be a function of positive type on a topological group G. Then, there exists a triple $(\pi, \mathscr{H}, \xi)$ consisting of a cyclic unitary representation with cyclic vector ξ such that

$$\varphi(g) = \langle \pi(g)\xi, \xi \rangle.$$

This triple is unique in the sense that if $(\rho, \mathscr{K}, \eta)$ is another such triple, then there exists an intertwining Hilbert space isomorphism $T : \mathscr{H} \to \mathscr{K}$ such that $T\xi = \eta$.

Proof. Let Φ be a kernel of positive type associated to φ. Then, there exists a Hilbert space $\mathscr{H}$ and a continuous map $f : G \to \mathscr{H}$ such that

$$\begin{aligned} \langle f(x), f(y) \rangle &= \Phi(x, y) \\ &= \varphi\left(y^{-1}x\right). \end{aligned}$$

We note that for every $g \in G$, $\langle f(gx), f(gy) \rangle = \langle f(x), f(y) \rangle$. Thus, by uniqueness, there is a unitary operator π_g on $\mathscr{H}$ such that $\pi_g f(x) = f(gx)$ for all $x \in G$. By density of $\{f(x) : x \in G\}$, we have that $\pi_{g_1 g_2} = \pi_{g_1}\pi_{g_2}$ for all $g_1, g_2 \in G$. Since f is continuous, $g \mapsto \pi_g \eta$ is continuous for every $\eta \in \mathscr{H}$. Set $\xi := f(e)$. Then, ξ is cyclic with $\varphi(g) = \left\langle \pi_g \xi, \xi \right\rangle$ for all $g \in G$.

If $(\rho, \mathscr{H}, \eta)$ is a triple with the same properties, then we have that

$$\Phi(g_1, g_2) = \left\langle \pi_{g_1}\eta, \pi_{g_2}\eta \right\rangle,$$

By uniqueness, there exists an isomorphism $T : \mathscr{H} \to \mathscr{K}$ such that $T\pi_g \xi = \pi_g \eta$ for all $g \in g$. Thus, $T\xi = \eta$, and T intertwines π and ρ since ξ is cyclic. □

Example: If G is a discrete group, and $\varphi = \delta_e$ is the Dirac mass at e, then the GNS representation for φ is the regular representation for G.

The Cone of Functions of Positive Type

We let $\mathcal{P}(G)$ be the set of functions of positive type on the topological group G. This forms a convex cone closed under complex conjugation and pointwise product. We let $\mathcal{P}_1(G)$ be the subset of $\varphi \in \mathcal{P}(G)$ such that $\varphi(e) = 1$.

Proposition: Let $\varphi_1, \varphi_2 \in \mathcal{P}(G)$, and let $\varphi = \varphi_1 + \varphi_2$. Then, $\pi_{\varphi_1} \leq \pi_\varphi$.

Moreover, if π_φ is irreducible, then $\varphi_1 = t\varphi$ for some $t \geq 0$.

References

[Con90] John B. Conway. *A Course in Functional Analysis*. Second. Vol. 96. Graduate Texts in Mathematics. Springer-Verlag, New York, 1990, pp. xvi+399. ISBN: 0-387-97245-5.

[BHV08] Bachir Bekka, Pierre de la Harpe, and Alain Valette. *Kazhdan's Property (T)*. Vol. 11. New Mathematical Monographs. Cambridge University Press, Cambridge, 2008, pp. xiv+472. ISBN: 978-0-521-88720-5. DOI: 10.1017/CBO9780511542749. URL: https://doi.org/10.1017/CBO9780511542749.